



Self-healing of vector field singularities

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ABSTRACT

We report here the self-healing ability of vector field singularities. While few studies discuss intensity healing in scalar beams, there is no report on polarization healing and healing of vector field singularities. Healing of polarization distributions from obstructions encountered by the beam is the high point of observation reported here as it entails healing of relative phases and amplitudes of the polarization component fields, in addition to intensity healing. This feature of robustness may play a significant role in applied research. The Poynting vector flow in the transverse plane of vector vortex beams is the key to the healing process. Since our investigation here is with vector field singularities, we also monitored the distribution of the Stokes parameter S_3 to confirm the healing process.

1. Introduction

Self-healing refers to the reconstruction of a partially obstructed beam upon propagation. Generally, this healing relates to the intensity healing of an optical beam [1,2]. This is an exciting and remarkable property as information, once lost, can be regained just through further propagation in free space. The intensity self-healing property has been noticed in several beams like Bessel [3–5], Airy [6], caustic [7], Mathieu [8], and conical beams [9]. This exciting self-healing property is useful in many fields like microscopy [4], laser micromachining [10], optical micromanipulation [11], and optical communication [12]. Laguerre–Bessel beams are observed to be more robust due to intensity self-healing than their peer Bessel beams [13,14]. The superposition of Laguerre–Gaussian beams also has the ability to self-reconstruct the intensity profile [2,15]. Further intensity self-healing has been shown in Bessel polarization singularities [16–19]. Self-healing of vector Bessel beams has also been shown and found application in quantum key distribution [20].

A vector optical field can be represented as a superposition of two orthogonal polarization component fields as [21],

$$\vec{E}(x, y) = a(x, y)e^{i\phi(x, y)} \begin{pmatrix} E_x(x, y) \\ E_y(x, y) \exp\{i\delta(x, y)\} \end{pmatrix}. \quad (1)$$

Where, $a(x, y)$ and $\phi(x, y)$ represent the common amplitude and phase distributions for both the components. $E_x(x, y)$ and $E_y(x, y)$ are real and normalized ($E_x(x, y)^2 + E_y(x, y)^2 = 1$). The polarization information is contained in the normalized Jones vector ($\hat{n}(x, y)$), i.e.,

$$\hat{n}(x, y) = \begin{pmatrix} E_x(x, y) \\ E_y(x, y) \exp\{i\delta(x, y)\} \end{pmatrix}. \quad (2)$$

$\delta(x, y)$ represents the phase retardation between the x and y components. For inhomogeneously polarized light, $\hat{n}(x, y)$ is not a constant. This means that at every spatial point, the state of polarization is different, and therefore $\hat{n}$ is written as $\hat{n}(x, y)$. The intensity distribution is decided only by amplitude $a(x, y)$; therefore, the phase and polarization are free parameters.

$$I(x, y) = [\vec{E}(x, y)] \cdot [\vec{E}(x, y)]^* = a^2(x, y) \quad (3)$$

Therefore, intensity healing reported in some beams may not witness complete phase or polarization healing. Since the state of polarization (SOP) of light is decided by the component field amplitudes and phases, the study of polarization healing assumes greater significance. For polarization healing, additional constraints such as the constant ratio of polarization component amplitudes and phases are demanded. In this article, the polarization healing accompanied by intensity healing of vector vortex beams subjected to obstructions is demonstrated for the first time.

2. Vector field singularities

Polarization singularities are structured optical beams that possess inhomogeneous polarization distribution. In these beams $\hat{n}(x, y)$ is spatially varying. The commonly used polarization attributes for polarization singularities are ellipticity χ and azimuth γ . These singularities are mainly classified as vector field singularities (V-points) and ellipse field singularities (C-points) [22–25]. V-points are intensity

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null points surrounded by linear SOPs with varying azimuths. C-points are points of circular polarization surrounded by elliptical SOPs with varying azimuths. Fields containing C-points have SOP distributions in which both ellipticity and azimuth are spatially varying. In V-points, the ellipticity across the beam remains constant, whereas azimuth is spatially varying. Fields containing both C-points and V-points are also there, but in these fields, constant ellipticity constraints on SOPs for V-points are applicable only locally. Further, in fields with inhomogeneous polarization distributions hosting some or none of these polarization singularities, the regions of the right and left-handed SOPs are separated by L – lines on which the SOPs are linear.

Vector field singularities are characterized by the Poincaré–Hopf index η defined by $\eta = \frac{1}{2\pi} \oint \nabla \gamma \cdot d\mathbf{l}$, where $d\mathbf{l}$ is a line element encircling the singularity. The state of polarization of light can be represented using Stokes parameters S_0, S_1, S_2 , and S_3 . For inhomogeneously polarized beams as the polarization vector $\hat{n}$ is spatially varying, i.e., function of (x, y) , all the Stokes parameters are also spatially varying. The Stokes field $S_{12}(x, y) = S_1(x, y) + i S_2(x, y) = A_{12} \exp(i \phi_{12}(x, y))$, generated using Stokes parameters $S_1(x, y)$ and $S_2(x, y)$ is used to identify these singularities. Since the S_1 and S_2 parameters contain azimuth information, the change in orientation of the state of polarization is reflected in the argument of the S_{12} field in the form of a phase vortex. The C-point and V-point singularities occurs at the intersection of zero contours of S_1 and S_2 . Inhomogeneously polarized beams can be represented as a single point on a hybrid order Poincaré sphere (HyOPS) [26] or a higher order Poincaré sphere (HOPS) [27]. Since the azimuth (γ) of a polarization ellipse is related to the Stokes parameter S_1 and S_2 as $\gamma = (1/2) \tan^{-1}(\frac{S_2}{S_1})$, the Stokes phase ϕ_{12} in terms of azimuth can be written as $\phi_{12} = 2\gamma$. The phase singularities in the Stokes field are related to polarization singularities [25]. At V-point singularity, all the Stokes parameters are zero, whereas at C-point, $S_1 = S_2 = 0$ and $S_3 = \pm 1$. For our study we consider vector field singularities with $\eta = +1$ and $\eta = +2$.

Polarization singularity being a superposition of orbital angular momentum (OAM) and spin angular momentum (SAM) states, increases the degrees of freedom in optical communication [28] and quantum information processing [29]. These beams have also found applications in beam shaping [30], robust beam engineering [31], chirality measurement [32], and image processing [33]. The polarization self-healing ability is a key feature that has not been studied so far. In this article, we make a detailed study of the self-healing property of the V-points, commonly referred to as vector vortex beams or vector field singularities. Interestingly, the obstructed intensity and polarization distribution are observed to self-heal in the far-field. We have demonstrated the self-healing of V-points using different types of obstacles: square, circular, and triangular.

A polarization singular beam can be mathematically expressed as [23]:

$$\vec{\psi}(r, \phi, z) = \psi_R \hat{R} + \psi_L \hat{L} \quad (4)$$

Where $\hat{R}$ and $\hat{L}$ are the circular basis vectors. $\psi_R = \alpha_R e^{i(l_R \phi + \Phi_R(z))}$ and $\psi_L = \alpha_L e^{i(l_L \phi + \Phi_L(z))}$ are the circular basis components of ψ . l_R and l_L are the topological charges of right and left circularly polarized vortex beams, respectively. α_R and α_L are the amplitude profiles of the two components. The phase term consists of an azimuthal phase factor of $\exp(i l_R \phi)$ and $\exp(i l_L \phi)$. Φ_R and Φ_L represent Gouy phases corresponding to vortex beams in the right and left circular polarization state respectively. The Gouy phases of the superposing scalar vortex beams are given by

$$\Phi_i(z) = (|l_i| + 1) \tan^{-1}\left(\frac{z}{z_R}\right); \quad i = R, L. \quad (5)$$

V-points are generated when the scalar vortices with topological charges of equal magnitude and opposite signs superpose, i.e., $l_R = -l_L$.

For a particular index $|\eta| = 1$, there are mainly four types of V-points: radial, anti-radial, azimuthal, and anti-azimuthal, which can be

directly produced from a laser cavity [24,34]. The intracavity generation methods of the radially polarized beam using binary dielectric grating [35], internal mirror [36], axicon [37], and Brewster elements [38,39] are also reported. Generation of the azimuthally polarized beam inside a Ruby Laser cavity [40] and high-power radially polarized beam in CO_2 lasers [41] are also shown. Moreover, the stable V-point singularities can be generated outside the laser cavity using metasurfaces [42], specially designed wave retarders [43], and interferometric methods [44]. These beams are also known as stable LP modes in optical fibers [45] namely TE_{01} (azimuthal), TM_{01} (radial), and HE_{21} (anti-radial and anti-azimuthal) and in cylindrical laser cavity resonators.

3. Healing of vector field singularities

Let $T(r, \phi)$ be the transmittance function of the obstacle. The field immediately after the obstruction is given by, $\vec{\psi}_O(r, \phi, z) = \vec{\psi}(r, \phi, z) \times T(r, \phi)$. This obstructed field is then propagated after a lens at various distances to observe the intensity and polarization distribution. The field $\vec{\psi}_O(r, \phi, z)$ can be written in Cartesian coordinate system as $\vec{\psi}_O(x, y, z)$. Let $\vec{E}_O(x, y, 0)$ be the field at the focal plane of a lens of focal length f . The angular spectrum of the field after propagating a distance z_1 from the minimum waist plane (focal plane) can be written as [46,47]:

$$\vec{E}_O(f_x, f_y; z_1) = \vec{E}_O(f_x, f_y; 0) \times e^{ikz\sqrt{1-\lambda^2(f_x^2+f_y^2)}} \quad (6)$$

where, (f_x, f_y) represent the spatial frequencies and λ is the wavelength.

$\vec{E}_O(f_x, f_y; 0)$ represent the 2D Fourier transform of the field at the focal plane of the lens. The field at plane z_1 is evaluated using inverse Fourier transforming the field $\vec{E}_O(f_x, f_y; z_1)$ as:

$$\vec{E}_O(x, y, z_1) = \mathcal{F}^{-1}\{\vec{E}_O(f_x, f_y; z_1)\}. \quad (7)$$

Here, symbol $\mathcal{F}^{-1}$ represents the inverse Fourier transform.

The Poynting vector has a magnitude of energy per second per unit area and a direction that represents the energy flow at any point in the field. This is the main factor responsible for the reconstruction of the blocked vortex beam. In an isotropic medium (here, free space), the Poynting vector is in the direction of the propagation. However, due to the helical wavefront of a vortex beam, the Poynting vector is also helical and advancing in the general beam propagation direction. It has components in the transverse plane and has a non-zero curl [48,49]. This transverse component helps in reconstructing the overall intensity structure of the blocked vortex beam after propagating a certain distance. The rotation (Θ) of the Poynting vector depends on the change in the Gouy phase during propagation and can be expressed as [50]:

$$\Theta = \frac{l}{2(|l| + 1)} \left(\frac{w(z)}{r(z)} \right)^2 \Phi(z). \quad (8)$$

Where $w(z)$ is the beam size, which depends on the minimum waist size w_0 and the Rayleigh distance z_R . Here, the distance z_R is measured from the focal plane of the lens. In the doughnut intensity structure of the beam, the rotation of the Poynting vector is shown [50] to depend on the magnitude and sign of the topological charge of the scalar vortex at radial distances where the intensity is not maximum. However, the amount of rotation is independent of the charge of the vortex for the radius corresponding to maximum amplitude [50,51].

Since the vector field singularities are superpositions of scalar vortices with charges of equal magnitude and opposite polarity, the Poynting vectors of each component vortex rotate in the opposite sense. The flow of energy in the transverse plane is termed internal energy flow [52,53]. It has been experimentally shown to be responsible for the intrusion of intensity into the obstructed region during the propagation of scalar vortices [49,51]. The transverse energy flow from both the

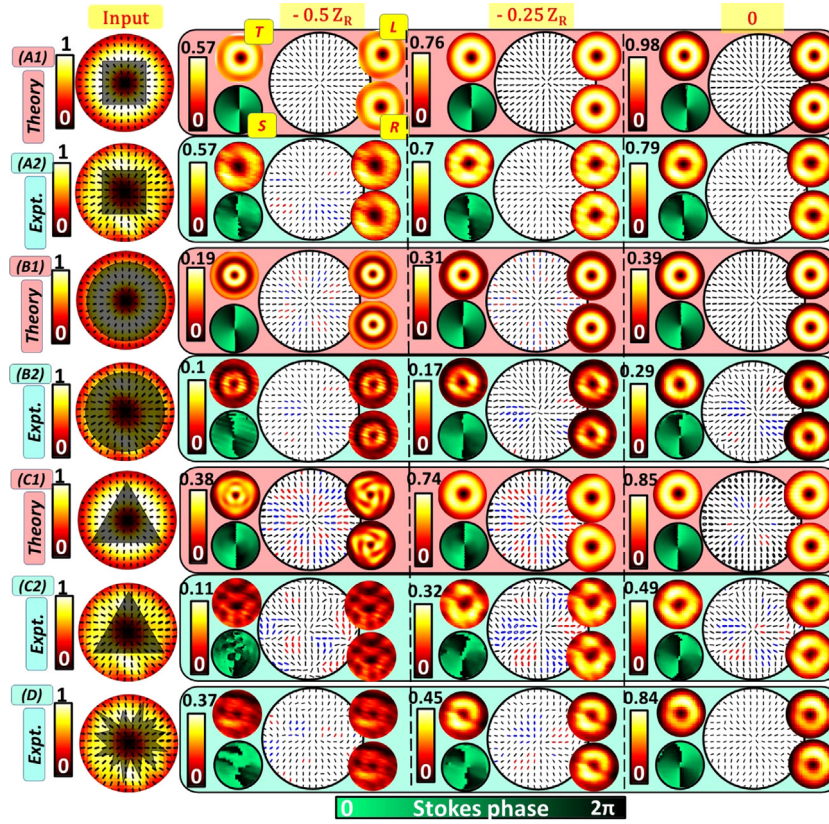


Fig. 1. Simulation (1) and experimental (2) results show self-healing of radially polarized beams when symmetrically blocked at the center by (A) square, (B) circular, (C) triangular, and (D) random obstruction. The polarization patterns of the obstructed beam are observed at different distances. The respective total intensity distributions (T), Stokes phases (S), LCP component (L), and RCP component (R) are represented for each SOP. The format for this representation is shown in the first case, followed all through the paper. Notably, all the obstructions are opaque but shown as semi-transparent here for clear visualization of the input beam.

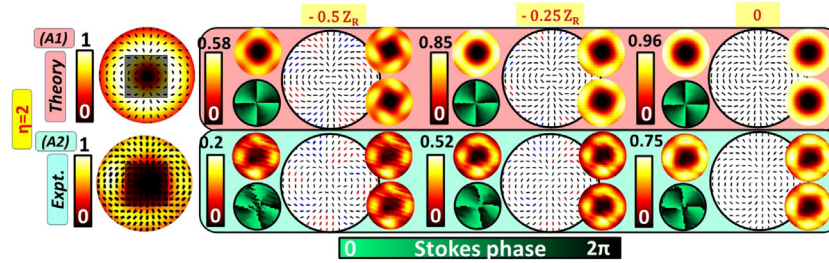


Fig. 2. Simulation and experimental results show a V point's self-healing with index $\eta = 2$ under a square obstacle. The polarization distribution of the obstructed beam at different distances is shown. The respective total intensity distributions, Stokes phases, LCP, and RCP components are also shown.

sides of the obstruction fills the darker region completing the intensity healing process. Also, the two components of the vector vortex beam superimpose to make the healing process faster as compared to a single scalar vortex. In scalar vortex beams, the role of phase vortices in intensity healing is amply demonstrated [9]. As far as polarization healing is concerned, for the vector vortex beams to maintain the polarization azimuth variation across the healed beam, the component phases in the circular polarization basis have to recover to maintain their respective helical phase variations. Further, to maintain zero ellipticity across the beam, the component amplitudes have to maintain a constant ratio. In other words, the right and left circularly polarized component scalar vortex beams exhibit complete recovery from obstruction during propagation.

4. Results and discussions

In this section, we present the simulation and experimental results to show the ability of vector vortex beams to self-heal both

intensity and polarization distribution after encountering different obstructions. Square, circular, and triangular obstacles are introduced one by one at the center of a radially polarized beam. The vector vortex beams reconstruct their intensity as well as polarization distributions, as depicted both theoretically and experimentally in Fig. 1. The total intensity distributions, Stokes phases, and polarization distribution of the obstructed beam at different distances are shown along with their respective circular polarization components: left circularly polarized (LCP), and right circularly polarized (RCP). On comparing the maximum total intensity regained after various obstructions, it can be visualized that the amount of intensity regained is maximum for a comparatively smaller amount of obstruction (for example, in Fig. 1 for square obstacle, theoretically 98% intensity is regained at the far-field). This is also quantified by calculating the correlation between the electric field with the obstacle $\vec{E}_O(x, y, z_1)$ and without the obstacle, $\vec{E}(x, y, z_1)$ at the far-field. The graph (Fig. 3) clearly shows that the value of the correlation decreases as the size of the obstacle (D) (side

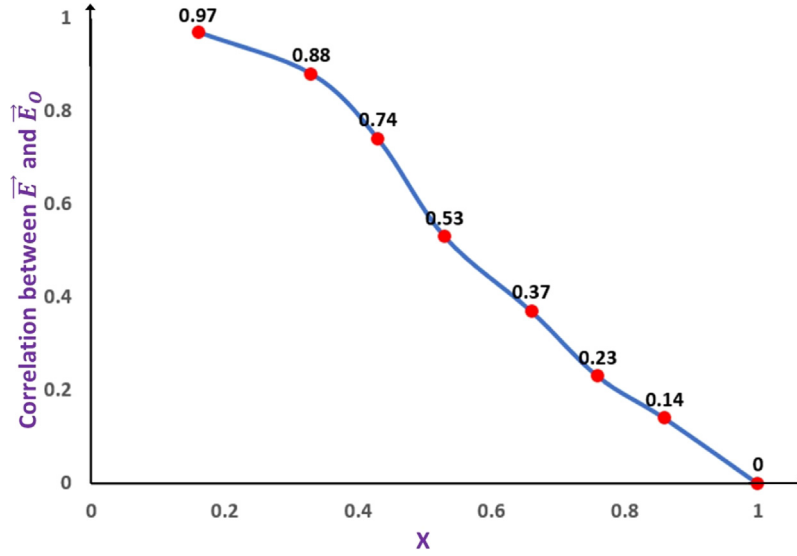


Fig. 3. Variation of correlation between obstructed ($\vec{E}_0$) and unobstructed beam ($\vec{E}$) in the far field with $X = D/w$ is shown. Here, D is the size of the obstacle (side length of square) and w is the beam size.

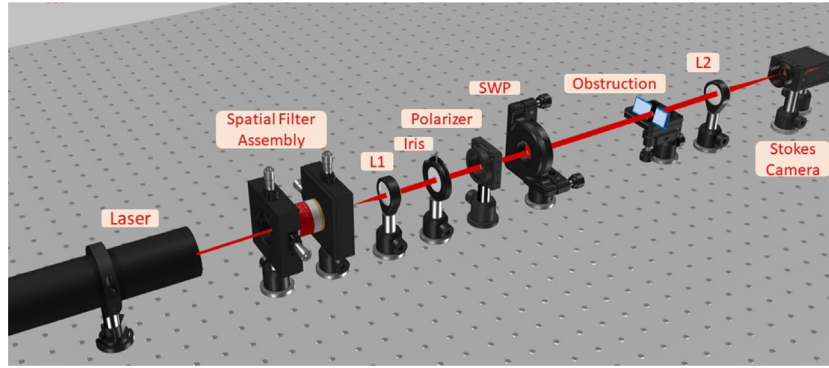


Fig. 4. Experimental setup for the observation of self healing. $L(1,2)$: Lens, SWP: S-wave plate, HWP: Half-wave plate.

length of the square) approaches the size of the beam (w). Therefore, the degree of obstruction decides the degree of self-healing.

The null points in the intensity distribution of RCP and LCP components can be seen to coincide in the far-field, assuring the equal amplitudes distribution and consequently the formation of vector vortex beams. The Stokes phase ϕ_{12} distribution is also observed to gain its appropriate phase distribution for the vector vortex beam in the far-field. Due to this recovery of relative phases and amplitudes of the polarization component fields, the polarization distribution in the vector vortex beams is observed to self-heal. Interestingly, both the intensity and the polarization distributions gain their initial structure at the far-field, making this property a promising one. Further, one can notice that the intensity healing precedes the polarization healing in a few cases (Fig. 1 B1, B2, and C1, C2). An example of self-healing of vector vortex beam with index $\eta = 2$ is also shown in Fig. 2. Complete self-healing at the far-field is not observed for obstructions that are de-centered.

The experimental setup to demonstrate the self-healing of vector field singularities is shown in Fig. 4. A He-Ne laser source of wavelength 632.8 nm is spatially filtered, collimated, and a beam with diameter 2 mm is obtained. It is then launched into a polarizer and an S-wave plate [43] (SWP) of charge $\eta/2$ for generating a V-point of index η . The generated V-point is further passed through different obstructions. This obstructed beam is then propagated to different distances. The resultant intensity distributions and Stokes parameters are recorded at three different planes using a Stokes camera (Salsa full Stokes polarization imaging camera, 1040×1040 pixels, BOSSA Nova, USA).

The experimental results to observe the self-healing of vector field singularities are also shown in Fig. 1 for different obstructions. A square of size $\approx 1.1 \text{ mm}$, a circle of radius $\approx 0.8 \text{ mm}$, an equilateral triangle of size $\approx 1.8 \text{ mm}$, and a random pattern are used as obstructions in the experiment. The experimental results are in good agreement with the simulated results. Since the V-points are constant ellipticity fields, the distribution of Stokes parameter $S_3(x, y)$ is also studied for V points (with index $\eta = 1$ and 2) obstructed by a square obstacle. It can be visualized from Fig. 5 that the fluctuations in the value of $S_3(x, y)$ are very low, thus, indicating the formation of V-point and its self-healing.

Thus, from the above discussions, it can be inferred that the vector vortex beams can self-heal on facing different obstructions. Vector vortex beams are known to be stable for communication in atmospheric turbulence [31]. However, there are reports that show that the cross-talk for scalar OAM modes and vector OAM modes is identical [54,55]. The polarization inhomogeneity is shown to be immune to perturbations that are unitary [56]. The self-healing property of vector singularities can be useful, especially in optical communication as the beam tries to reconstruct any lost information during propagation.

5. Conclusion

In conclusion, in this article, the ability of vector vortex beams to self-heal both intensity and their polarization distribution under various obstructions is presented for the first time. Recovery of polarization distribution is possible because of the recovery of relative phases and

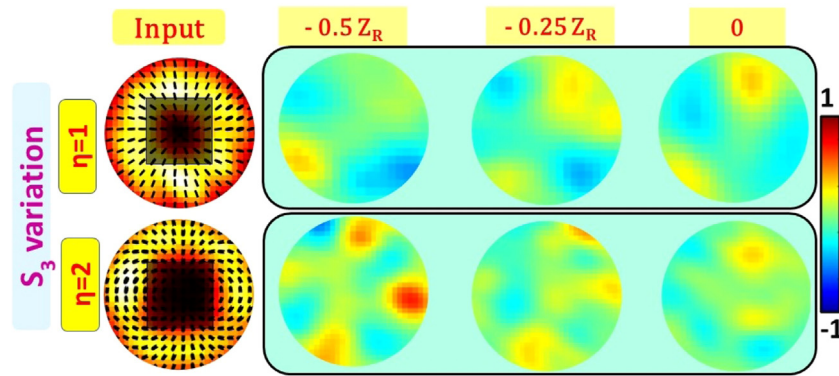


Fig. 5. Experimental distribution of Stokes parameter S_3 at three different z planes.

amplitudes of the polarization component fields. The Poynting vector flow in the transverse plane of these beams plays an important role in the healing process. This self-reconstruction property of vector vortex beams can be utilized for applications in different fields like optical communication, microscopy, and optical micromanipulation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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